

Experiment 2

Preparation of a Software Project Plan Covering Scope, Timeline, and Team Roles for an AI Study Assistant

Aim

To prepare a software project plan for an AI Study Assistant by defining its scope, timeline, and team responsibilities.

Objective

- I) Define the project scope.
- II) Identify functional requirements.
- III) Prepare a project timeline.
- IV) Assign team roles.
- V) Plan project resources and risks.

1. Project Overview

The AI Study Assistant is a web-based application that helps students learn efficiently using AI. It allows users to upload notes, ask academic questions, generate quizzes, and monitor their study progress.

- ★ Target Users
- ★ Students
- ★ Teachers
- ★ Educational Institutions

2. Project Scope

- In Scope
- User Registration & Login
- AI Chat Assistant
- PDF Upload
- Quiz Generation
- Progress Tracking
- Notifications
- Out of Scope
- Video Classes
- Online Payments
- Course Marketplace

3. Functional Requirements

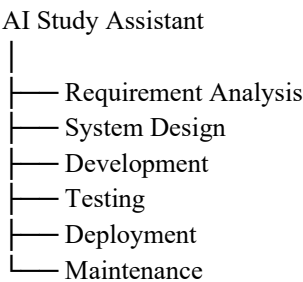
- Student Module
- Login
- Upload Notes
- AI Chat
- Generate Quiz
- View Progress

- AI Module
- Answer Questions
- Generate Summaries
- Recommend Study Plans
- Admin Module
- Manage Users
- Generate Reports
- Monitor System

4. Non-Functional Requirements

- 1. Fast Response Time
- 2. Secure User Authentication
- 3. Easy-to-use Interface
- 4. Reliable System
- 5. Scalable Architecture

5. Work Breakdown Structure



6. Project Timeline

Phase	Duration
Requirement Analysis	Week 1
Design	Week 2
Development	Week 3–7
Testing	Week 8–9
Deployment	Week 10
Documentation	Week 11–12

7. Team Roles

Role	Responsibility
Project Manager	Project Planning
Business Analyst	Requirement Gathering
UI Designer	Interface Design
Frontend Developer	User Interface
Backend Developer	APIs & Database
AI Engineer	AI Integration
Tester	Testing
DevOps Engineer	Deployment

8. Resource Plan

- i. Software
- ii. VS Code
- iii. GitHub
- iv. Figma
- v. Postman
- vi. Technology
- vii. React
- viii. Python (Fast API)
- ix. PostgreSQL
- x. AWS

9. Risk Management

Risk	Solution
AI Errors	Improve prompts and testing
Development Delay	Weekly sprint planning
Security Issues	User authentication and encryption
Server Failure	Cloud backup

10. Milestones

- ❖ Requirement Analysis Completed
- ❖ Design Approved
- ❖ Development Completed
- ❖ Testing Completed
- ❖ Application Deployed

11. Expected Outcome

The AI Study Assistant will provide students with AI-powered learning support, quiz generation, and progress tracking through a secure and user-friendly platform.

12. Conclusion

The project plan provides a structured approach for developing the AI Study Assistant by defining the project scope, schedule, team roles, and resources. Proper planning helps complete the project efficiently and ensures successful delivery